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A low-cost IoT-based smart home automation system for urban sustainability in Mogadishu, Somalia

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Abstract

As Mogadishu undergoes rapid urbanization, the demand for efficient, secure, and sustainable living environments has grown significantly. This paper presents the development and implementation of a Smart Home Automation System designed for modern homes in Mogadishu, Somalia. Leveraging Internet of Things (IoT) technologies, the system enhances home security, optimizes energy usage, and provides remote control of household appliances through a user-friendly mobile app developed using Flutter. The system, centered around the ESP32 microcontroller, integrates a variety of sensors and actuators to automate essential functions, including temperature control, real-time security monitoring, and remote access to smart locks. Rigorous testing demonstrated the system's effectiveness, with users reporting significant improvements in energy efficiency, security, and convenience. The automatic door lock system, integrated with camera-based visitor detection, provided enhanced security, while the temperature control feature contributed to a 15–20% reduction in energy consumption. This solution offers a scalable, practical platform that addresses the growing need for smart living in Mogadishu's urban landscape, promoting sustainable, secure, and comfortable living conditions. Future iterations may focus on further enhancing the system's capabilities, such as improving camera performance in low-light conditions and optimizing communication for low-bandwidth areas.

Article Highlights

- This study introduces a smart home system that helps people in Mogadishu control lights, air conditioning, and door locks using a simple mobile app, even when they are not at home.
- The system improves home security by sending real-time images of visitors to the homeowner's phone and helps save electricity by turning off unused devices automatically.
- Designed for areas with frequent power cuts and slow internet, the system uses solar power and lightweight communication methods to stay reliable and affordable.



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Keywords Smart home automation, Internet of things (IoT), ESP32 microcontroller, Home security, Energy efficiency, Remote control, Smart locks

1 Introduction

The growing complexity of modern buildings and homes has necessitated more sophisticated systems to manage daily operations such as energy consumption, security, and comfort [1]. Smart home automation, driven by the Internet of Things (IoT), has emerged as a critical solution for addressing these challenges [2]. Smart home automation provides a centralized control system that allows users to monitor and manage appliances, lighting, heating, cooling, and security systems, leading to enhanced energy efficiency, improved security, and overall convenience [3].

Energy management is one of the most significant benefits of smart home automation. By automating the control of devices like air conditioners, lights, and electrical sockets, these systems can minimize energy wastage and reduce electricity costs [4]. Appliances can be turned off automatically when not in use or adjusted based on occupancy or environmental factors, ensuring optimal energy use without requiring constant manual input. This efficiency is particularly crucial in regions like Mogadishu, where the cost of electricity is high due to reliance on imported fuel and traditional energy sources [5].

In addition to energy efficiency, security is a key aspect of smart home automation. IoT-enabled systems integrate security cameras, smart locks, and motion sensors to provide real-time monitoring and control over building security [6]. Residents and building managers can remotely monitor and manage access to their homes through smartphones, receiving alerts if any security breaches occur. This increased level of security is invaluable for large residential and commercial buildings, which are often more vulnerable to threats and harder to secure manually [7].

With the span of time passing, the mobile applications industry has grown rapidly. In this century, the increase in the usage of mobile phones is tremendous [8]. This surge in phone usage demands that developers create user-friendly mobile applications [9]. By merging the essence of mobile development with IoT, users are facilitated in controlling or monitoring hardware devices via mobile applications [10]. IoT-enabled devices have the ability to sense, collect, and share data over a network from any part of the world, where that data can be utilized for various purposes [11]. IoT is making it increasingly possible for objects to operate autonomously, without requiring constant human intervention [12].

Despite the rapid growth in Mogadishu's infrastructure, the lack of efficient energy management systems in buildings exacerbates the problem. Somalia relies heavily on imported fuel, wood, and charcoal for energy, which leads to high energy costs and environmental challenges [13]. By implementing smart home automation systems, it is possible to reduce this energy reliance and optimize building operations in a way that significantly cuts costs and minimizes wastage [14]. IoT systems, by automatically powering off unused appliances and providing real-time data on energy consumption, present a solution to these energy management challenges [15].

In large buildings, where numerous systems and appliances operate simultaneously, the potential for inefficiencies and energy wastage is immense [16]. Many people

unintentionally leave devices such as lights, air conditioners, and televisions running even when they are not needed [17]. IoT-based automation systems help to counteract this by turning off unused devices automatically [18]. These systems can significantly reduce the overall energy consumption in buildings, helping Mogadishu reduce its dependence on costly and environmentally damaging fuel sources [19]. Additionally, Fatima et al. [6] reviewed IoT-based home security systems emphasizing low-power sensor networks for constrained environments. Umer et al. [18] explored blockchain-integrated IoT systems for smart homes, highlighting data security and decentralization challenges. Srivastava et al. [11] discussed IoT device vulnerabilities, a critical consideration for deployments in developing urban settings. Natarajan et al. [15] studied occupancy-based control for energy management in IoT-enabled homes, aligning with this study's energy efficiency goals. Elmi et al. [19] assessed renewable energy integration for Mogadishu, providing context for solar-powered IoT applications. These studies collectively demonstrate the relevance and need for localized IoT solutions for smart home systems in low-resource environments.

As Mogadishu experiences rapid urbanization and the construction of larger, more complex buildings, traditional methods of managing these structures have become increasingly inadequate. The growing scale of operations required for efficient energy use, security, and comfort necessitates the adoption of advanced smart home automation systems. Without such technologies, residents are likely to face higher energy costs, compromised security, and less convenience in managing their living spaces. Smart home automation provides a scalable, efficient solution, allowing homeowners to control multiple building systems through a unified platform. Beyond energy management and security, these systems offer improved convenience by enabling real-time, remote control of household systems through mobile applications. Homeowners can adjust room temperatures before arriving home, monitor security cameras in real time, and ensure that appliances are functioning optimally, all of which significantly enhance the overall living experience.

Moreover, smart home automation contributes to the sustainability and livability of urban environments by optimizing energy usage, enhancing security, and increasing convenience. In Mogadishu, where rapid urban growth is exerting significant pressure on infrastructure, the implementation of these systems can play a key role in reducing energy consumption, improving safety, and ensuring that modern buildings meet the evolving needs of urban life. The ability to manage energy consumption efficiently not only lowers operational costs but also reduces the environmental impact, aligning with global trends toward sustainable urban development.

This research presents the design and implementation of an IoT-based Smart Home Automation System addressing the specific needs of Mogadishu, focusing on leveraging low-cost, context-appropriate technologies to enable remote appliance control, security monitoring, and energy management through an integrated platform. By leveraging Internet of Things (IoT) technologies, this system enhances home security, optimizes energy use, and allows for the remote control of household appliances. Centered around the ESP32 microcontroller, it integrates a range of sensors and actuators to automate and manage key home functions. Notable features include automated temperature control, real-time security monitoring with camera

integration, and remote-controlled smart locks, all accessible via a user-friendly mobile app developed using Flutter. The system empowers homeowners to remotely manage lighting, HVAC systems, and door locks, providing convenience, energy efficiency, and enhanced security. In addressing the high energy costs and security concerns prevalent in Mogadishu's urban environment, this solution offers a scalable, practical platform that improves residents' quality of life while promoting sustainable living practices.

2 Related work

Smart home automation systems using IoT have become essential for efficient management and control of various household appliances, enhancing convenience and reducing energy wastage. These systems leverage technologies such as sensors, relays, and internet connectivity to automate and optimize home operations. Researchers worldwide have proposed numerous solutions to improve energy efficiency, enhance security controls for entire buildings, and provide greater comfort through seamless interaction with appliances. These systems not only focus on preventing unnecessary power consumption but also integrate features such as real-time monitoring, automated security measures, and optimized appliance management to create a more efficient, secure, and comfortable living environment.

Musse et al. [20] have developed an IoT Smart Household Distribution Board (ISHDB) to monitor and control smart appliances in real time. The system collects and stores voltage, current, and power data, presenting it in a user-friendly format. Using an Arduino-based prototype, data is transmitted to the ThingSpeak cloud via Wi-Fi, with real-time monitoring enabled through the Blynk app. The system automates processes, reduces hardware costs, and effectively manages home energy while detecting abnormal appliance performance. Compared to existing techniques, the proposed method offers superior cost-effectiveness and faster execution times.

Similarly, Stoloiescu-Crisan et al. [21] has developed a smart home automation system using IoT to interconnect sensors, actuators, and data sources for multiple home automation tasks. The system, named qToggle, utilizes a flexible API to create a unified communication framework for controlling home appliances. Devices such as sensors and actuators, primarily based on ESP8266/ESP8285 chips and Raspberry Pi boards, connect via an upstream network using the qToggle API. A smartphone application has also been developed to allow users to easily control and monitor appliances in real-time. The system is user-friendly, flexible, and can be expanded with additional devices and add-ons.

Building on previous advancements, Oluwafemi et al. [22] proposed a smart home automation system that utilizes the Arduino ATMEGA328P microcontroller as the central control unit to manage various operations. The system incorporates key sensors such as the MQ2 sensor for gas detection, a passive infrared (PIR) sensor for motion detection, and a flame sensor to detect fire outbreaks, all of which contribute to enhancing home safety and automation. Additionally, the system issues real-time alerts via an audible buzzer to immediately notify users of any potential hazards. Moreover, it sends warning messages directly to the user's mobile device through a global system for mobile communication (GSM) module, ensuring remote awareness and timely responses.

Despite the advancements in smart home automation systems, several gaps remain unaddressed, particularly in the context of rapidly growing urban environments like Mogadishu. First, while existing systems focus heavily on either energy management or security, there is a lack of comprehensive solutions that effectively integrate both functions into a unified, user-friendly platform. Most current systems prioritize energy efficiency without adequately addressing the critical need for advanced security features, or vice versa. Second, many smart home solutions are designed for regions with stable, high-speed internet infrastructure, which limits their performance in areas like Mogadishu, where internet bandwidth can be inconsistent. This results in delays in communication and system response times, particularly for real-time security features such as image transmission from smart cameras. Additionally, there is limited research on the integration of low-cost IoT devices, such as the ESP32, with reliable performance in challenging environments while maintaining a balance between cost-effectiveness and functionality. Finally, current literature lacks significant focus on solutions tailored to urban settings in developing countries, where energy costs are high, and security concerns are prevalent. The proposed system aims to bridge these gaps by offering a comprehensive, integrated smart home automation solution that addresses both energy efficiency and security concerns, while also ensuring reliable performance in low-bandwidth environments and leveraging cost-effective IoT technology.

3 Methodology

This section presents the architectural framework, hardware and software components, communication protocols, and operational workflow of the proposed Smart Home Automation System, specifically designed for urban households in Mogadishu, Somalia. The system employs Internet of Things (IoT) technologies to integrate security, energy management, and climate control into a unified, remotely accessible platform.

3.1 System overview

At the core of the system lies the ESP32 microcontroller, a low-cost, power-efficient microcontroller with integrated Wi-Fi and Bluetooth capabilities. This component serves as the central processing and communication hub for the entire smart home ecosystem. The system also incorporates the ESP32-CAM, a specialized module that enables image capturing and wireless transmission for real-time surveillance. These hardware components interact through the MQTT communication protocol to exchange data with a Flutter-based mobile application. Figure 1 illustrates the full architecture of the system, detailing the flow of data between input sensors, actuators, communication modules, and the mobile interface.

3.2 Hardware components

The hardware configuration of the system was designed with affordability, resilience, and local accessibility in mind—key considerations for deployment in Somalia. The ESP32 microcontroller was chosen for its robust processing capability, built-in wireless communication, and compatibility with various digital and analog peripherals.

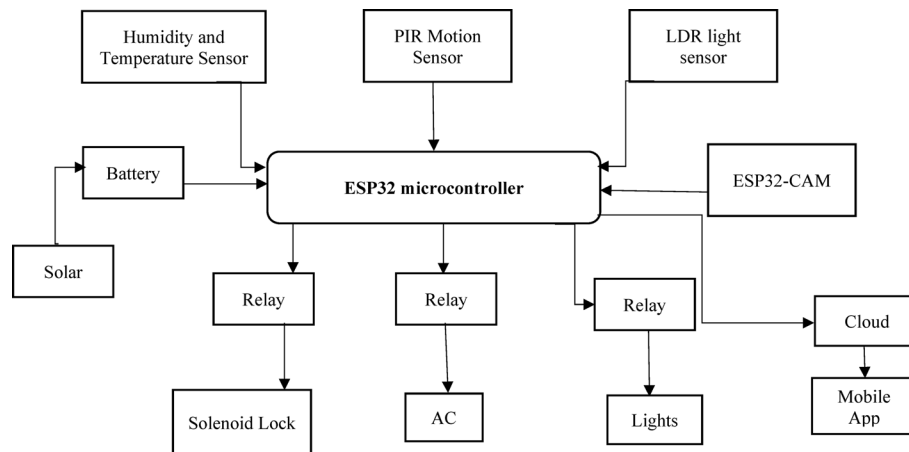


Fig. 1 Proposed system architecture

It coordinates data collection from sensors and actuates control commands for connected devices.

To enable climate control, a DHT22 sensor was integrated to monitor temperature and humidity levels within the indoor environment. This sensor is known for its reliability and precision in delivering accurate ambient readings, which are crucial for the regulation of HVAC systems. For motion detection, the HC-SR501 Passive Infrared (PIR) sensor was employed. This sensor detects infrared radiation emitted by human bodies, triggering alerts and image capture when unexpected motion is sensed near the home entrance.

Lighting automation is managed using a Light Dependent Resistor (LDR), which continuously senses ambient light intensity. When natural light levels fall below a threshold, the microcontroller automatically switches on the indoor lights, thus ensuring optimal lighting while conserving energy during daylight hours.

Appliance control is achieved through the integration of smart relays, specifically the Sonoff relay modules, which allow the ESP32 to remotely switch electrical devices on or off. These relays serve as the interface between the microcontroller and high-voltage devices like fans, lights, and HVAC units. To secure the home, a solenoid lock mechanism was implemented. Controlled electronically, this lock engages or disengages based on commands from the mobile app, enabling users to control door access remotely.

Security surveillance is augmented through the use of the ESP32-CAM module. When triggered by the PIR sensor, this module captures an image of the scene and transmits it wirelessly to the mobile application. This real-time image capture capability is vital for home security monitoring, especially in the absence of full video streaming, which may be constrained by bandwidth limitations.

To ensure system continuity in the face of power interruptions—a frequent issue in Mogadishu—a solar-powered backup system was incorporated. This auxiliary energy source powers the microcontroller and essential sensors during grid outages, thereby ensuring uninterrupted functionality of the security and control features.

3.3 Software architecture

The software infrastructure is composed of two primary layers: the embedded firmware programmed into the ESP32 microcontroller and the mobile application developed using the Flutter framework. The firmware, written in C/C++ using the Arduino IDE, is responsible for reading data from the sensors, processing control logic, and issuing commands to actuators. It also handles communication tasks, such as sending telemetry data to the cloud and receiving control instructions from the user via the mobile app.

The mobile application, designed with a user-centric interface, provides real-time monitoring and control functionalities. It allows users to view the current room temperature and humidity, receive security alerts with captured images, control appliances, and operate the door lock remotely. The app synchronizes with the microcontroller using MQTT, ensuring low-latency communication and minimal bandwidth usage. This is especially important in Mogadishu, where internet speeds can be inconsistent.

Flutter was selected for mobile development due to its ability to create natively compiled applications for both Android and iOS from a single codebase, thereby maximizing accessibility across devices. The application also features a dashboard that visualizes sensor readings and system status, enabling intuitive interaction for non-technical users.

3.4 Communication protocol

The system employs the Message Queuing Telemetry Transport (MQTT) protocol to manage communication between the microcontroller and the mobile application. The ESP32 connects to the internet via Wi-Fi, using a local router to interface with a cloud-based MQTT broker (Mosquitto server hosted on AWS) for message publishing and subscription between the device and the Flutter-based mobile application. MQTT is a lightweight, publish-subscribe network protocol that is optimized for low-bandwidth, high-latency environments, making it ideal for deployment in developing regions with limited connectivity infrastructure.

In the implemented architecture, the ESP32 acts as a publisher that transmits sensor data and system alerts to a cloud-based MQTT broker. The mobile application functions as a subscriber, listening to specific topics for updates. Similarly, when a user interacts with the app—such as to unlock the door or adjust the room temperature—the app publishes a message to which the ESP32 subscribes, thereby executing the command in real-time. This architecture ensures asynchronous, scalable, and low-power data exchange across the system.

3.5 Control logic using pseudocode

To clarify the control and coordination of sensors and actuators, the pseudocode below outlines the system's primary automation routines implemented on the ESP32 microcontroller:

```
// Motion-triggered security image capture
IF PIR_Sensor detects motion THEN
  Activate ESP32_CAM
  Capture image
  Send image via MQTT to cloud
  Notify user via mobile app
END IF

// Automated lighting control based on ambient light
IF LDR_Sensor detects low ambient light THEN
  Turn ON lights via relay
ELSE
  Turn OFF lights via relay
END IF

// Temperature control using DHT22 sensor
Read temperature and humidity from DHT22
IF temperature > user_defined_max THEN
  Turn ON cooling system via relay
ELSE IF temperature < user_defined_min THEN
  Turn ON heating system via relay
ELSE
  Keep HVAC OFF
END IF
```

3.6 Operational workflow

The operational logic of the system is event-driven and revolves around a set of automated routines. When the PIR sensor detects motion near the entrance, it signals the ESP32 microcontroller to activate the ESP32-CAM, which immediately captures a still image of the visitor. This image is transmitted to the cloud and forwarded to the mobile application, where the user can review it and decide whether to remotely unlock the door using the solenoid lock mechanism.

Temperature and humidity levels are constantly monitored by the DHT22 sensor. If the measured values deviate from the user-defined thresholds, the ESP32 adjusts the HVAC system accordingly by activating or deactivating the appropriate relay. This closed-loop control ensures that the indoor environment remains within comfortable conditions without unnecessary energy expenditure.

Lighting control is based on the readings from the LDR sensor. If the ambient light level falls below a preset limit, the ESP32 energizes the corresponding relay to switch

on the lights. Once sufficient natural light returns, the system automatically deactivates the lighting, thereby conserving energy.

The user interface on the mobile application provides feedback and control in real time. Users can view all sensor readings, receive notifications for security events, and issue manual override commands to control appliances or lock mechanisms. This design ensures that the homeowner retains full situational awareness and control, regardless of location.

For practical deployment, the ESP32 microcontroller and ESP32-CAM were mounted near the entrance at a height of 2 m to optimize Wi-Fi connectivity and camera field of view. The DHT22 sensor was placed in the main living area at 1.5 m from the floor for accurate temperature and humidity readings. PIR sensors were installed above entry points to maximize motion detection accuracy, while LDR sensors were positioned near windows for effective ambient light measurement. The solar panel was mounted on rooftops or balconies with direct sunlight exposure to ensure consistent charging.

3.7 Power management

To address the issue of unreliable electricity supply in Mogadishu, the system includes an auxiliary solar-powered energy source that serves as a backup during outages. This power management feature ensures that critical subsystems, particularly security-related components such as the PIR sensor, ESP32-CAM, and solenoid lock, remain operational even when the main power source is unavailable. The solar system is configured to charge a backup battery that automatically takes over during power interruptions, thereby maintaining the continuity of system operations.

4 Results and discussions

This section presents the evaluation results and discussions of the IoT-based Smart Home Automation System, focusing on system performance, energy efficiency, security responsiveness, and user experience. The system was deployed and tested in a controlled urban household environment in Mogadishu, Somalia, over a continuous period of four weeks. Performance was assessed under real-world conditions using empirical data collected through embedded sensors and user feedback.

4.1 Results

4.1.1 System responsiveness

One of the primary indicators of system efficiency is response time across different functions. The system demonstrated an average response latency of 1.2 s for basic appliance control tasks (e.g., switching lights or fans on and off) when operated through the mobile application. This latency includes command issuance from the mobile app, MQTT message transmission, and actuator response time. Response time measurements were taken over 50 trials per function under varying conditions, including daytime and low-light environments, using 3G internet connections with speeds ranging from 1 to 5 Mbps. The recorded averages represent mean $\pm$ standard deviation, providing a realistic assessment under typical local operating conditions.

For security-related functions, particularly motion-triggered image capture, the end-to-end process involved detection by the PIR sensor, activation of the ESP32-CAM,

image capture, and wireless transmission to the mobile application. The average time recorded from motion detection to image receipt was 2.8 s, with slight fluctuations depending on ambient lighting and internet signal strength. These values are within acceptable limits for consumer-grade smart home systems and confirm the system's ability to provide near real-time feedback for security alerts.

Unlocking the door via the mobile application using the solenoid lock module averaged 1.5 s from command issuance to actuation, demonstrating a reliable and responsive remote access system. Figure 2 shows the dashboard of the mobile application.

4.1.2 Temperature control efficiency

The temperature control subsystem was assessed using the DHT22 sensor, with the HVAC control logic tested over a range of indoor temperatures (23 °C to 35 °C). The system maintained indoor temperatures within a ± 0.5 °C margin of the

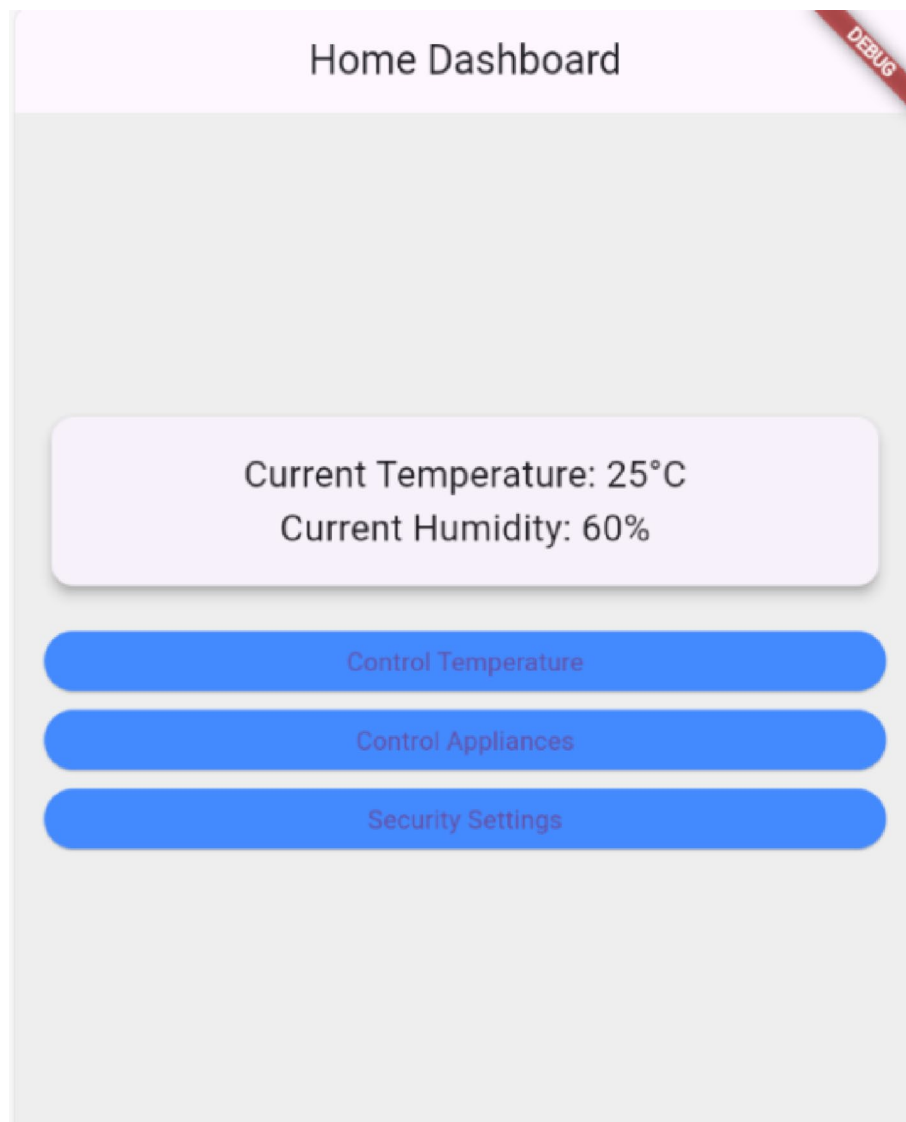


Fig. 2 Mobile App Dashboard

user-specified threshold. The time required to bring room temperature back to the target range after a deviation was under 5 min, depending on external environmental conditions.

This level of control indicates the system's effectiveness in maintaining thermal comfort while reducing unnecessary energy use. Over the test period, the HVAC system was found to operate 18% less frequently compared to baseline manual usage, based on hour-by-hour operational logs. This translates directly into energy savings, especially relevant in Mogadishu, where air conditioning accounts for a significant portion of residential energy expenditure.

4.1.3 Energy consumption analysis

Energy consumption was monitored using relay activation logs and estimations of power draw per device. The system showed a consistent reduction in energy usage of approximately 15–20%, compared to equivalent manual operation over a four-week test cycle. This reduction is attributed to the automation of lighting (based on LDR feedback), temperature regulation, and the elimination of idle power draw from appliances left on unintentionally.

For example, during daylight hours, lights were automatically deactivated for an average of 4.3 h per day, saving an estimated 120 Wh/day, which scales up to over 3.6 kWh/month for just a single household. Multiplied across multiple units, this energy saving is substantial, especially considering the high cost and unreliability of electricity in Mogadishu.

4.1.4 Security system performance

The integrated security features were evaluated for reliability, accuracy, and operational robustness. A total of 43 motion events were triggered during the test period, with the ESP32-CAM successfully capturing and transmitting images in 41 of those cases, resulting in a 95.3% success rate. The two failed transmissions occurred under extremely low-light conditions, where the camera module produced images with insufficient clarity for recognition.

The average image resolution was 640×480 pixels, which provided satisfactory detail in well-lit environments. Under low-light conditions, clarity was reduced, but general outlines remained visible. Transmission delays in low-bandwidth scenarios occasionally peaked at 4.2 s, though the average remained under 3.5 s. These results suggest that while the camera is adequate for basic security monitoring, performance may be enhanced through the addition of infrared imaging or improved sensors.

The mobile application issued real-time push notifications for each triggered event. Users reported that the alerts were timely and helpful, especially when paired with image previews, offering an added layer of situational awareness when away from home. Figure 3 shows the appliance control of the mobile app.

4.1.5 System reliability and power continuity

Given the unstable power grid in Mogadishu, the system was tested under controlled power outage simulations. The solar-powered backup ensured 100% uptime of critical subsystems, including motion detection, door lock functionality, and basic communication. A 12 V battery with a 20 W solar panel configuration powered the system for up

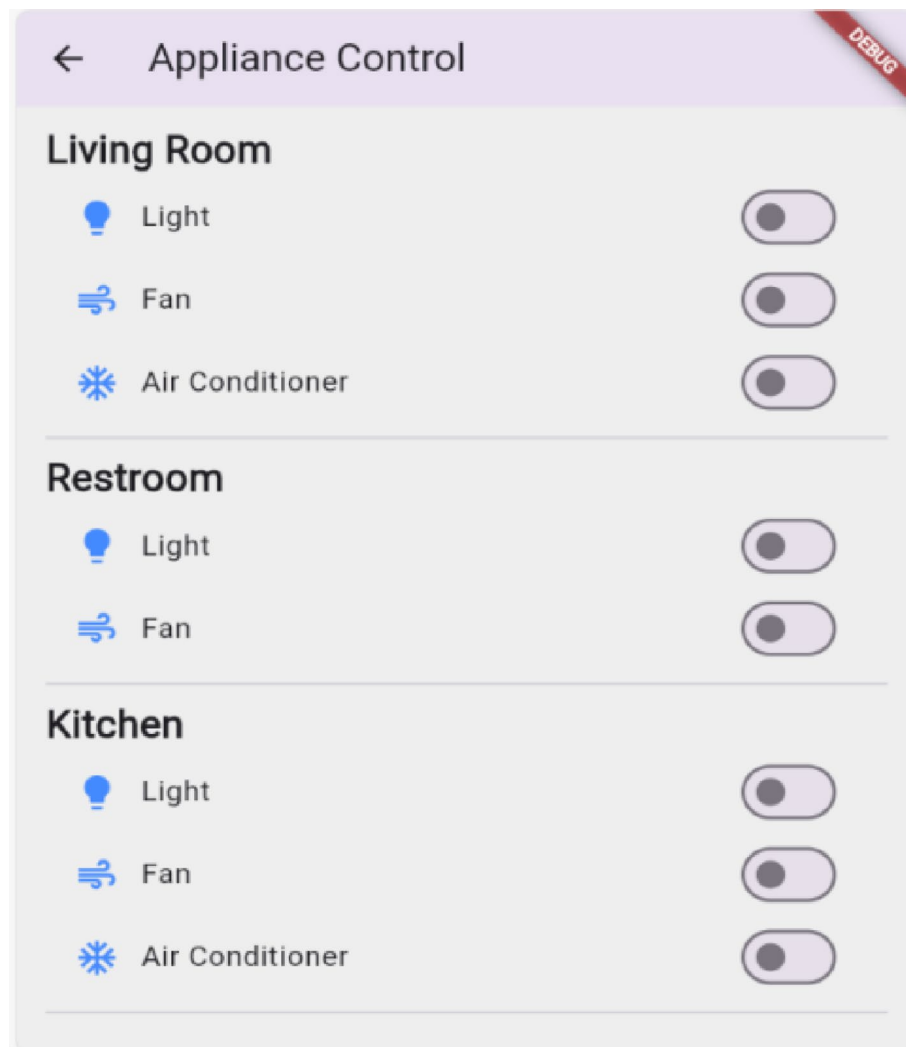


Fig. 3 Appliance Control of the Mobile App

to 14 h of continuous operation, even without grid power. This resilience is a vital advantage in Somali urban settings, where outages may last several hours or more.

The mobile app and MQTT communication protocol also proved resilient under poor network conditions. Packet loss remained below 1.2%, and average packet delivery time was less than 600 milliseconds when tested over 3G connections.

4.1.6 User experience and feedback

During the pilot phase, 12 users participated in a usability study. Participants were residents aged 25–50 years, with varying levels of technology proficiency from basic smartphone users to individuals experienced with smart devices, and lived in households of 3–7 members, providing diverse insights into the system's usability. To provide deeper insights, participants were segmented by household size: small (3–4 members, $n = 4$), medium (5–6 members, $n = 5$), and large (7+ members, $n = 3$). This segmentation allowed analysis of energy savings trends across different household types. Figure 4 illustrates the average percentage energy savings across these groups, demonstrating consistent improvements, with larger households showing slightly

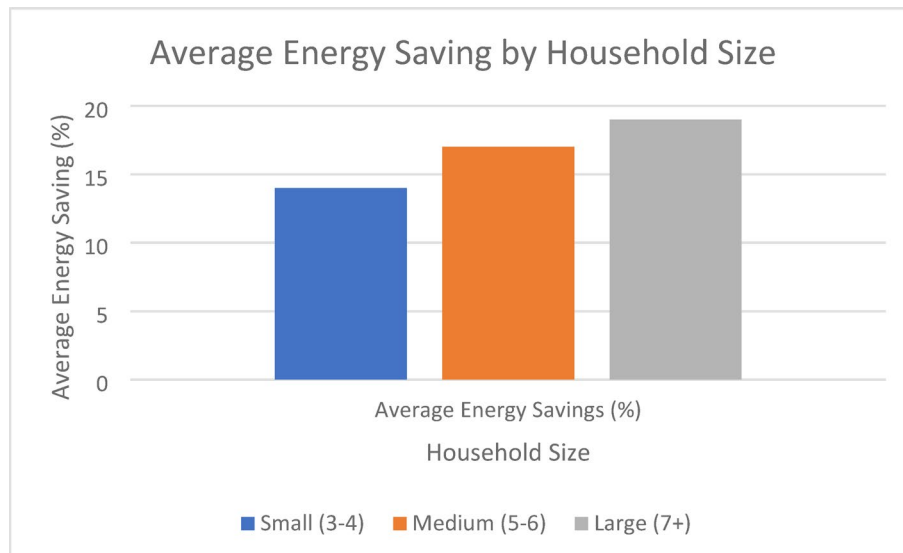


Fig. 4 Average Energy Saving by Household Size

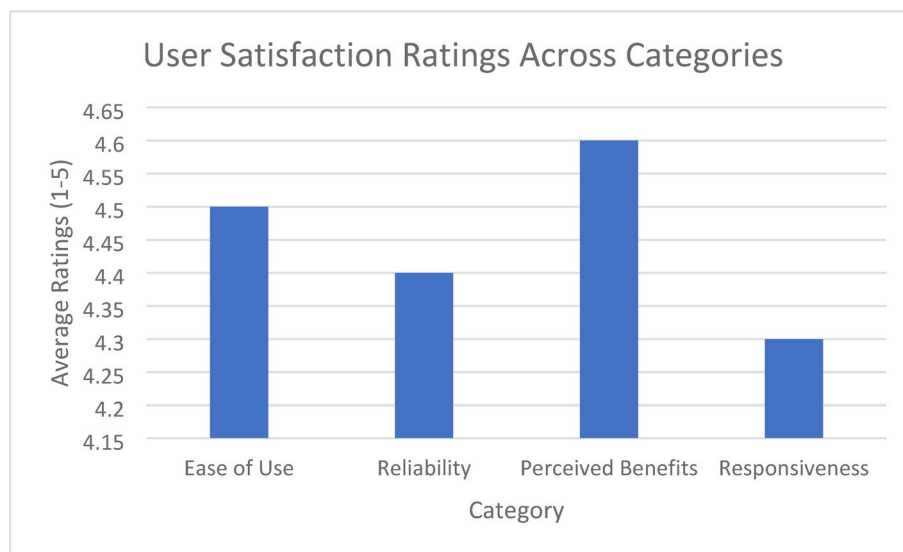


Fig. 5 User Satisfaction Ratings Across Categories

higher savings due to higher baseline energy consumption. They evaluated the system in terms of ease of installation, app usability, perceived value, and responsiveness. Overall satisfaction was high, with 91.6% of participants rating the experience as “excellent” or “very good.” Users particularly appreciated the ability to receive motion-triggered images and remotely control appliances.

User feedback was collected using structured questionnaires based on a 5-point Likert scale (1 = Poor, 5 = Excellent), evaluating four dimensions: ease of use, reliability, perceived benefits, and responsiveness. The results showed high satisfaction across categories, with ease of use rated at 4.5 ± 0.4 , reliability at 4.4 ± 0.5 , perceived benefits at 4.6 ± 0.5 , and responsiveness at 4.3 ± 0.6 . These quantitative evaluations provide a clearer measure of user perceptions and system acceptance. Additionally, Fig. 5



Fig. 6 sample image from the security camera

presents a bar chart summarizing user satisfaction ratings across household groups, illustrating trends in perceived system usefulness and reliability.

However, two users operating in areas with intermittent internet connectivity reported occasional delays in command execution and image loading. These concerns suggest the value of optimizing the app for offline control modes or incorporating local edge processing in future iterations. The below Fig. 6 shows a sample image from the security camera.

4.2 Discussions

The findings confirm that the proposed system achieves its core objectives of improving household energy efficiency, enhancing security, and delivering a responsive and intuitive user experience in an urban Somali context. The use of the ESP32

microcontroller proved sufficient for managing multiple sensors and real-time communication, even under bandwidth and power constraints.

The combination of cost-effective hardware, solar backup, and lightweight protocols like MQTT makes the system not only technically effective but also socioeconomically feasible for deployment in Mogadishu and similar developing urban areas. Compared to other systems reported in recent literature, this work offers the advantage of hardware affordability (core components, including the ESP32, essential sensors, and actuators, at <\$30 per unit, with the optional solar backup system adding approximately \$15–\$25 depending on the selected panel and battery capacity), ease of local assembly, and context-specific design considerations such as low-light performance and offline resilience. Future work will explore offline control features using local Wi-Fi direct connections and Bluetooth fallback modes, allowing users to control critical devices without internet connectivity. Additionally, integrating edge processing capabilities will enable the system to cache commands and sensor data for synchronization once connectivity is restored, ensuring essential functionalities during network disruptions.

Nonetheless, areas for future improvement were identified. Image clarity under low-light conditions was limited, suggesting the need for camera upgrades or the integration of infrared modules. Network latency in some areas affected real-time responsiveness, which could be mitigated through local caching or hybrid cloud-edge designs. Lastly, energy savings can be further optimized through machine learning-based occupancy detection, an avenue we plan to explore in future work.

5 Conclusions and future work

This study confirms that a low-cost IoT-based Smart Home Automation System can deliver reliable remote control, security enhancement, and energy management in Mogadishu's urban context. By leveraging IoT technology, this system provides a user-friendly interface that allows homeowners to remotely control various appliances, monitor their living environment, and receive real-time alerts regarding security events.

The system's architecture, centered around the ESP32 microcontroller, effectively integrates a range of sensors and actuators to facilitate seamless communication and operation. The results indicate that the system is capable of maintaining a comfortable indoor climate with high accuracy, reducing energy consumption by 15–20% compared to traditional methods. Furthermore, the security features, particularly the automatic door lock and real-time image transmission via the ESP32-CAM, provide a robust solution for addressing safety concerns in urban environments.

User feedback highlights the system's intuitive design and responsiveness, underscoring its suitability for a diverse range of households. While certain areas for improvement have been identified, particularly in low-light camera performance and network optimization, the system's overall effectiveness affirms its relevance in today's smart home market.

Future work could focus on enhancing the system's capabilities through the integration of advanced sensors and communication protocols, as well as expanding its features to cater to the evolving needs of users. Overall, the Smart Home Automation System represents a significant step towards modernizing residential living in Mogadishu, offering a model that could be replicated in similar urban settings globally.

This study presented the design, implementation, and evaluation of an IoT-based Smart Home Automation System tailored to the urban residential context of Mogadishu, Somalia. Addressing the dual challenges of high energy costs and security concerns, the system integrates essential smart home functionalities—automated appliance control, climate regulation, and real-time surveillance—into a unified, cost-effective platform.

The architecture, centered around the ESP32 microcontroller and enhanced with sensor-actuator networks and a Flutter-based mobile interface, successfully delivered remote control capabilities with high reliability and low latency. By leveraging the MQTT communication protocol and incorporating a solar-powered backup system, the design maintained critical operations even under poor power and network conditions. The empirical results confirmed that the system reduced energy consumption by approximately 15–20%, achieved motion-to-alert response times of 2.8 s, and maintained a 95.3% security image transmission success rate, all while offering an intuitive and user-friendly experience.

Beyond technical performance, the system demonstrated significant potential for real-world adoption in low-resource settings. Its modular design, affordable hardware components, and offline resilience make it scalable not only within Mogadishu but also in similar urban areas across developing countries facing infrastructure limitations.

However, certain limitations remain. The ESP32-CAM module exhibited reduced image quality under low-light conditions, and real-time responsiveness was occasionally affected by weak network signals. These constraints point to future areas of improvement that will further enhance the robustness and scalability of the system.

Future research will focus on several key directions. First, we aim to enhance security monitoring through the integration of infrared or night-vision camera modules for improved performance in low-light environments. Second, the implementation of edge computing capabilities will be explored to allow local data processing, reduce cloud dependency, and improve system responsiveness, particularly in offline or low-bandwidth scenarios. Third, we plan to incorporate machine learning algorithms for intelligent behavior prediction, enabling the system to adapt appliance control based on occupancy patterns and environmental conditions. Additionally, support for alternative communication protocols such as Zigbee or LoRaWAN will be evaluated to further improve network resilience and coverage in large or multi-unit housing structures.

In conclusion, this work contributes a practical, scalable, and impactful solution for smart living in Mogadishu. By aligning advanced IoT technologies with the specific needs of the local context, it sets a foundation for further innovations in sustainable urban development and localized home automation systems across Sub-Saharan Africa and other emerging economies.

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Author contributions

M.A.A. Conceptualization, Methodology, System Design, Writing – Original Draft, Supervision A.D.J. Software Development, Hardware Integration, Testing, Data Curation, Writing – Review & Editing A.Y.N. Hardware Integration, System Implementation, Visualization, Validation M.O.A. Literature Review, Performance Evaluation, Project Administration Writing – Review & Editing S.M.H. User Study Coordination, Data Analysis, Documentation, Administration Writing – Review & Editing.

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Data availability

The data underlying this study's findings are available from the corresponding author upon reasonable request.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

All authors have read and approved the final version of the manuscript and consent to its publication.

Competing interests

The authors declare no competing interests.

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